

# **Aesthetic Physique Gym Nutrition Plan: A Logistical and Metabolic Integration Strategy for Extreme-Budget Hostel Environments**

## **Executive Summary of Metabolic and Logistical Parameters**

The pursuit of body recomposition—the simultaneous oxidation of adipose tissue alongside the accretion of lean muscle mass—demands precise macronutrient partitioning, an exact caloric deficit, and an elevated protein intake to stimulate muscle protein synthesis (MPS). Executing this physiological target within the infrastructure of a traditional Indian college hostel environment introduces severe socio-economic and logistical bottlenecks. The subject of this analysis is a 24-year-old male student residing in Kolkata, West Bengal, presenting with a stature of 169 cm and a body mass of 68 kg. The metabolic directive is a rigid 2,050 kilocalorie (kcal) recomposition deficit, derived directly from the "Aesthetic Physique Gym Nutrition Plan." The logistical framework governing this intervention is highly restricted. The subject relies upon a fixed ₹100 per day hostel mess subscription, which yields unmeasured quantities of cooked white rice, highly diluted lentil soup colloquially termed "watery dal," and strictly apportioned animal proteins consisting of 50 to 75 grams of chicken at 11:00 AM and 50 to 75 grams of fish at 8:00 PM. Culinary infrastructure is non-existent, limited exclusively to a standard electric water kettle. Furthermore, the subject operates under an absolute supplementary nutrition budget of exactly ₹1,500 per month, translating to a daily liquidity limit of merely ₹50. This document provides an exhaustive, data-driven framework designed to bridge the chasm between extreme financial limitation and advanced sports nutrition. It demonstrates the mathematical viability of procuring a high-biologic-value "Kettle Stack" (comprising whole eggs, textured soya chunks, roasted Bengal gram sattu, and raw peanuts) for under ₹50 daily within the current Kolkata retail and wholesale markets. Furthermore, it mathematically derives the exact allowance of mess-provided white rice required to anchor the 2,050 kcal target, explores the thermodynamics of kettle-based food preparation, and provides an alternative matrix of ultra-cheap, kettle-friendly substitutes to mitigate dietary fatigue over extended macrocycles.

## **Anthropometric Profiling and Institutional Dietary Constraints**

The subject's anthropometrics—24 years of age, 169 cm in height, and 68 kg in mass—indicate a Body Mass Index (BMI) of approximately 23.8. While this falls within the standard normal range, aesthetic bodybuilding and body recomposition protocols rely on the manipulation of lean tissue versus adipose tissue, rendering BMI a secondary metric. The prescribed caloric intake of 2,050 kcal signifies a moderate deficit engineered to encourage lipolysis while providing

sufficient energy and essential amino acids to stimulate hypertrophy following resistance training.

To achieve muscular hypertrophy within a caloric deficit, dietary protein must be elevated significantly above the baseline Recommended Dietary Allowance (RDA). Current sports nutrition literature dictates an intake of 1.6 to 2.2 grams of protein per kilogram of body weight for resistance-trained individuals. For a 68 kg subject, this necessitates a daily protein yield of 109 grams to 150 grams. Dietary fats must be maintained at approximately 0.7 grams to 1.0 gram per kilogram of body weight (47 grams to 68 grams) to ensure optimal endocrine function, specifically the biosynthesis of steroidal hormones. The remaining caloric allowance is strictly allocated to carbohydrates to replenish hepatic and muscular glycogen stores and sustain high-intensity anaerobic output.

## **The Nutritional Reality of the Indian Hostel Mess**

Institutional hostel diets in India present a unique metabolic challenge, predominantly biased toward high-glycemic carbohydrates and low-quality fats, accompanied by severe deficiencies in bioavailable protein. The unlimited availability of white rice provides cheap, rapid-digesting glucose, but the accompanying "watery dal" offers negligible macronutrient value. Institutional dal is typically subjected to extreme dilution to maximize volume and reduce overhead costs, rendering its protein and caloric content fractional compared to traditional home-cooked preparations.

The inclusion of animal protein—specifically 50 to 75 grams of chicken and 50 to 75 grams of fish daily—is a significant advantage over standard vegetarian mess plans. However, these specific institutional portions are mathematically insufficient to meet the athlete's 109-gram minimum daily protein requirement. Therefore, the ₹50 per day supplementary budget must be engineered with absolute precision to source the cheapest, highest-yielding protein variants available in the Kolkata metropolitan area, necessitating zero advanced cooking infrastructure.

## **Financial Engineering: The ₹50 Daily Supplementary Kettle Stack**

The core of this nutritional intervention relies entirely on the "Kettle Stack": 4 whole eggs, 50 grams of textured soya chunks, 50 grams of chana sattu, and 30 grams of raw peanuts. Procuring this highly specific combination within a rigid ₹50 daily limit is challenging but empirically viable when leveraging wholesale market dynamics, local supply chains, and bulk purchasing strategies within Kolkata.

## **Market Pricing Dynamics in Kolkata (July 2026)**

To validate the financial feasibility of the Kettle Stack, retail and wholesale price indices across Kolkata must be scrutinized against current market data.

**Whole Eggs Procurement** The National Egg Coordination Committee (NECC) wholesale rate for eggs in Kolkata serves as the baseline for pricing. As of July 2026, the NECC rate fluctuates consistently between ₹7.00 and ₹7.20 per piece. While local retail markups often push single-egg purchases to ₹7.50 or higher in convenience stores, the strict budget constraint dictates that the subject must bypass retail margins by purchasing standard 30-egg trays. A 30-egg tray in Kolkata's wholesale hubs, such as the Sealdah or Maniktala markets, or through

local discount grocers, costs exactly ₹213.00 to ₹216.00. The egg market in Kolkata is highly volume-dependent, with bulk buyers securing a "peti" (210 eggs) for approximately ₹1,491 to ₹1,512, though a 30-egg tray is the most practical volume for a hostel resident. The procurement strategy mandates the purchase of one 30-egg tray every 7.5 days.

**Textured Soya Chunks Procurement** Textured Vegetable Protein (TVP), commercially known as soya chunks, offers the highest protein-to-rupee ratio available in the Indian domestic market. While premium branded cartons (such as Nutrela 200g packs) cost approximately ₹45 (translating to ₹225 per kg), bulk purchasing significantly circumvents this premium. Loose soya chunks or local institutional bulk packs (such as those from local Kolkata distributors like ZKP or Sparsh) retail between ₹60 and ₹120 per kilogram. The strategy dictates that the student purchases a standard 1-kilogram bag monthly from a local wholesale grocer at a highly conservative estimate of ₹100 per kilogram.

**Chana Sattu Procurement** Sattu, the dry-roasted and stone-ground flour of black Bengal gram (kala chana), is an indigenous dietary staple in Eastern India. It requires zero cooking, rendering it the ultimate functional food for a hostel environment. In Kolkata, established commercial brands like Ganesh Chana Sattu dominate the retail space. A 500-gram pouch of Ganesh Chana Sattu is priced between ₹70.10 and ₹84.00, depending on the retailer and bulk discounts. Loose, unbranded organic chana sattu can be sourced directly from local mills for as low as ₹80 to ₹100 per kilogram. To ensure quality, purity, and hygiene, this financial model relies on purchasing a reliable local brand like Ganesh or high-quality loose sattu at an average of ₹140 per kilogram.

**Raw Peanuts Procurement** Raw peanuts supply the primary lipid anchor for the supplementary stack, offering essential monounsaturated fats. In Kolkata's wholesale and local retail markets, bold raw groundnuts range from ₹60 to ₹120 per kilogram depending on the grade and harvest season. Packaged corporate varieties can reach up to ₹195 per kilogram, but loose procurement is absolutely mandatory for budget adherence. The financial model assumes a conservative loose retail price of ₹100 per kilogram, necessitating a monthly purchase of roughly one kilogram.

## Cost Verification and Economic Viability Matrix

| Dietary Component              | Daily Prescribed Quantity | Current Kolkata Rate (per Kg / Bulk Unit) | Calculated Daily Cost (₹) |
|--------------------------------|---------------------------|-------------------------------------------|---------------------------|
| Whole Eggs (Commercial Layer)  | 4 Pieces                  | ₹216.00 per 30-piece tray                 | ₹28.80                    |
| Textured Soya Chunks           | 50 Grams                  | ₹100.00 per kilogram                      | ₹5.00                     |
| Roasted Chana Sattu            | 50 Grams                  | ₹140.00 per kilogram                      | ₹7.00                     |
| Raw Peanuts (Kernels)          | 30 Grams                  | ₹100.00 per kilogram                      | ₹3.00                     |
| <b>Total Daily Expenditure</b> | -                         | -                                         | <b>₹43.80</b>             |

The mathematical total of ₹43.80 proves unequivocally that the Kettle Stack is economically viable. This framework leaves a daily buffer of ₹6.20, equating to approximately ₹186 per month. This surplus capital must be strategically hoarded to purchase essential micronutrients and condiments, such as iodized salt, black pepper, lemons (to enhance sattu iron absorption), or instant coffee sachets for pre-workout central nervous system stimulation.

## Biochemical Profiling and Physiological Impact of the

# Kettle Stack

Understanding the biochemical composition and physiological impact of these four specific ingredients is critical for maximizing body recomposition and ensuring that the ₹43.80 daily expenditure yields the highest possible return on investment for muscle protein synthesis.

## The Role of Whole Eggs

Eggs represent the gold standard for protein bioavailability, scoring a perfect 100 on the Biological Value (BV) scale and serving as the reference point for all other dietary proteins. Four large commercial layer eggs provide approximately 25.2 grams of highly bioavailable, complete protein. This protein is exceptionally rich in leucine, the primary branched-chain amino acid (BCAA) responsible for stimulating the mammalian target of rapamycin (mTOR) pathway, the primary driver of muscle hypertrophy. Furthermore, the 19 grams of dietary fat derived exclusively from the yolks is rich in dietary cholesterol. In the context of a healthy, resistance-trained male, dietary cholesterol serves as a direct precursor molecule for the biosynthesis of testosterone and other critical androgenic hormones necessary for tissue repair. The yolks additionally supply choline, a micronutrient essential for optimal central nervous system function, neurotransmitter synthesis, and hepatic lipid metabolism.

## The Role of Textured Soya Chunks

Textured soy protein is a highly concentrated, defatted soy flour derivative. A modest 50-gram serving yields an immense 26 grams of protein. While plant-derived proteins are frequently critiqued in sports nutrition for presenting incomplete amino acid profiles, soy stands as a rare exception. It possesses a Protein Digestibility Corrected Amino Acid Score (PDCAAS) approaching 1.0, containing all nine essential amino acids. When integrated into a diet alongside the methionine-rich eggs and the mess-provided chicken and fish, the overall amino acid pool is perfectly complemented. Additionally, soya chunks demonstrate a high satiety index; they absorb massive quantities of water during rehydration, aiding in mechanical gastric distension. This physiological stretch response suppresses the secretion of ghrelin (the hunger hormone), an invaluable effect during a sustained caloric deficit.

## The Role of Chana Sattu

According to the Indian Council of Medical Research (ICMR) and the National Institute of Nutrition (NIN) 2017 Indian Food Composition Tables, 100 grams of chana sattu delivers roughly 20 to 22.5 grams of protein, 65 grams of complex carbohydrates, and 7.6 grams of dietary fiber. A 50-gram daily serving provides approximately 12.6 grams of protein and 32.3 grams of slow-releasing carbohydrates. Brand-specific data for Ganesh Chana Sattu confirms a similar macronutrient yield.

Crucially, the traditional dry-roasting process pre-gelatinizes the starches and degrades anti-nutritional factors such as phytic acid, rendering the flour instantly digestible and highly bioavailable without the need for cooking. Sattu presents a very low glycemic index (GI) of approximately 40. This low GI metric prevents rapid insulin spikes, promoting highly stable blood glucose levels and sustained ATP generation. The presence of resistant starch within the sattu also functions as a prebiotic, bypassing the small intestine to ferment in the colon, thereby

producing short-chain fatty acids that reduce systemic inflammation and support gut microbiome integrity. Furthermore, sattu is highly concentrated in micronutrients, with a 100-gram serving providing up to 800 mg of potassium, 166 mg of magnesium, and 8 to 9 mg of iron.

## The Role of Raw Peanuts

Peanuts act as the primary caloric and lipid anchor for the supplementary stack. A 30-gram serving delivers approximately 170 kcal, 7.7 grams of protein, and 14.8 grams of fat. The lipid profile of the peanut is dominated by oleic acid (a monounsaturated fat) and linoleic acid (a polyunsaturated fat). Consuming these dense, slow-digesting fats in proximity to the high-glycemic mess rice fundamentally alters the gastric emptying rate of the total meal. By delaying the absorption of glucose into the bloodstream, the peanuts effectively lower the overall glycemic load of the diet, mitigating the post-prandial insulin crashes that frequently trigger lethargy and acute hunger in deficit states.

## Integration with Institutional Mess Offerings

The fixed ₹100 per day mess subscription forms the caloric foundation of the diet. The nutritional variables provided by the mess are standard white rice, watery dal, chicken, and fish. Analyzing the macronutrient density of these items is critical for determining the final daily calorie count.

### Analysis of the Institutional Mess Proteins

**11:00 AM Meal: Chicken Curry (50-75g)** Assuming a median serving of 62.5 grams of cooked chicken, typically served in a spiced gravy. The cuts in institutional settings are often bone-in and may include skin. However, isolating the edible meat portion, 62.5 grams of standard Indian cooked chicken curry yields approximately 100 kcal, 11.3 grams of protein, and 5.6 grams of fat.

**8:00 PM Meal: Fish Curry (50-75g)** In the context of Kolkata, institutional fish is almost exclusively Rohu (*Labeo rohita*) or Katla, traditionally prepared as *Macher Jhol* (a light curry cooked in mustard oil with turmeric). Rohu is an exceptionally lean, high-quality freshwater fish. Assuming a median edible yield of 62.5 grams, it provides roughly 81.3 kcal, 9.4 grams of protein, and 4.4 grams of fat. Rohu is particularly valuable for sports nutrition due to its omega-3 fatty acid content (specifically EPA and DHA), which downregulates systemic inflammation caused by the mechanical damage of heavy resistance training and accelerates muscular recovery. Comparatively, Rohu contains lower total fat and calories than Katla, making it vastly superior for strict weight loss and recomposition phases.

### Analysis of the "Watery Dal"

Institutional dal is stretched with excessive water volumes to minimize lentil consumption per capita. While raw lentils contain roughly 24 grams of protein per 100 grams, a standard 150ml *katori* of highly diluted hostel dal may contain as little as 10 to 15 grams of actual cooked lentils. The caloric and macronutrient yield is therefore negligible. Assuming the consumption of two bowls (300ml total) per day, the nutritional contribution is estimated at merely 80 kcal, 3.0 grams of protein, 0.4 grams of fat, and 16.0 grams of carbohydrates. Despite its low nutrient density, the dal is logistically valuable for hydration and providing a palatable fluid medium to mix with

the dry mess rice and the rehydrated soya chunks.

## The Rice Calculation Engine: Deriving the Caloric Anchor

With the Kettle Stack and the mess proteins mathematically quantified, the sole remaining variable is the precise volume of cooked white rice required to reach the 2,050 kcal target.

### Step 1: Quantifying the Fixed Caloric Inputs

The combined macronutrient and caloric yield of the fixed daily items must be aggregated to determine the remaining allowance for the rice component.

| Dietary Component         | Exact Daily Quantity | Energy (kcal)  | Protein (g) | Fat (g)     | Carbs (g)   |
|---------------------------|----------------------|----------------|-------------|-------------|-------------|
| Whole Eggs                | 4 Pieces (Large)     | 286.0          | 25.2        | 19.0        | 1.4         |
| Soya Chunks               | 50g (Raw Weight)     | 172.5          | 26.0        | 0.3         | 16.5        |
| Chana Sattu               | 50g (Dry Weight)     | 196.9          | 12.6        | 3.0         | 32.3        |
| Raw Peanuts               | 30g (Raw Weight)     | 170.1          | 7.7         | 14.8        | 4.8         |
| Chicken (Mess)            | 62.5g (Cooked Meat)  | 100.0          | 11.3        | 5.6         | 1.3         |
| Fish (Mess)               | 62.5g (Cooked Meat)  | 81.3           | 9.4         | 4.4         | 1.3         |
| Watery Dal                | 300ml (2 Bowls)      | 80.0           | 3.0         | 0.4         | 16.0        |
| <b>Total Fixed Matrix</b> | -                    | <b>1,086.8</b> | <b>95.2</b> | <b>47.5</b> | <b>73.6</b> |

*Data derived from computational nutritional models based on standard Indian food composition.*

### Step 2: Isolating the Rice Allowance

The target caloric intake is strict at 2,050 kcal. Subtracting the fixed matrix calories reveals the remaining budget.

- **Target Calories:** 2,050.0 kcal
- **Fixed Matrix Calories:** 1,086.8 kcal
- **Remaining Caloric Allowance:**  $2,050.0 - 1,086.8 = 963.2$  kcal

This remaining 963.2 kcal must be derived entirely from the hostel's cooked white rice.

### Step 3: Converting Calories to Grams and Volumetric Hostel Bowls

The energy density of cooked white rice is significantly lower than that of raw rice due to massive water absorption during the boiling process. While 100 grams of raw white rice contains approximately 356 kcal to 365 kcal, 100 grams of plain cooked white rice contains only 130 kcal.

This is a frequent point of failure in amateur dietary tracking.

To determine the exact weight in grams of cooked white rice required to fill the 963.2 kcal deficit, the following formula is applied:

- $963.2 \text{ kcal} \div 1.30 \text{ kcal/g} = \mathbf{740.9 \text{ grams of cooked white rice.}}$

In a standard Indian institutional setting, digital kitchen scales are rarely accessible in the dining hall. Therefore, the metric weight must be translated into reliable volumetric units. In India, a standard stainless steel *katori* (small bowl) holds between 130 grams and 150 grams of cooked rice when filled to the brim without severe mechanical packing. Using the 150-gram standard:

- $740.9 \text{ grams} \div 150 \text{ grams/bowl} = \mathbf{4.94 \text{ standard bowls.}}$

For rigorous practical application, the subject must consume exactly **5 standard hostel bowls** (katoris) of cooked white rice per day. It must be noted that "bowl deception" is common; if the hostel provides a larger 200-250g serving bowl, the subject must adjust accordingly. Assuming the standard 150g katori, this rice volume should be partitioned between the two primary mess meals: 2.5 bowls during the 11:00 AM chicken meal, and 2.5 bowls during the 8:00 PM fish meal.

## Final Daily Macronutrient Summation

Integrating the 740.9 grams of cooked white rice—which yields approximately 20 grams of trace plant protein, 2.2 grams of trace fat, and 207 grams of carbohydrates—into the daily matrix results in the following comprehensive daily macronutrient profile:

| Total Daily Macros         | Absolute Value      | Relative to Body Weight (68 kg) |
|----------------------------|---------------------|---------------------------------|
| <b>Total Calories</b>      | <b>~2,050 kcal</b>  | <b>30.1 kcal / kg</b>           |
| <b>Total Protein</b>       | <b>~115.1 grams</b> | <b>1.69 grams / kg</b>          |
| <b>Total Fats</b>          | <b>~49.7 grams</b>  | <b>0.73 grams / kg</b>          |
| <b>Total Carbohydrates</b> | <b>~281.0 grams</b> | <b>4.13 grams / kg</b>          |

*Data derived from aggregate nutritional analysis.*

This macronutrient split (approximately 22% Protein, 22% Fat, 56% Carbohydrate) is optimally configured for a recomposition phase. It provides sufficient glucose to fuel intense anaerobic resistance training, adequate lipids for hormonal stability, and a robust 1.69g/kg of protein to ensure maximal stimulation of muscle protein synthesis while in a net energy deficit.

## Thermodynamics and Logistical Execution of Kettle-Based Preparation

The absolute reliance on a solitary electric kettle for food preparation requires specific thermodynamic protocols to ensure food safety, nutrient retention, and the longevity of the appliance itself.

### Protocol 1: Hard-Boiling Eggs via Passive Thermal Transfer

Electric kettles utilize a bi-metal switch that automatically terminates the electrical circuit when the steam pressure indicates the water has reached 100°C. Standard stovetop boiling instructions, which rely on sustained active boiling, are therefore inapplicable.

1. **Placement:** Gently place 4 room-temperature eggs at the bottom of the empty kettle. Eggs removed directly from refrigeration are highly susceptible to thermal shock and will

likely fracture upon contact with rapidly heating water.

2. **Submersion:** Pour cold water over the eggs until they are submerged by a minimum of 2.5 centimeters (1 inch). Overfilling risks violent overflow through the spout; underfilling results in uneven protein coagulation.
3. **Thermal Activation:** Activate the kettle. Allow it to reach a full rolling boil and automatically disengage.
4. **Passive Cooking Phase:** Do not breach the thermal seal by opening the lid. Leave the eggs in the highly insulated hot water for exactly 12 to 13 minutes. This passive residual heat provides a gentle carry-over cooking process, fully solidifying the yolk without producing the grey-green ferrous sulfide ring associated with aggressive, rapid overcooking.
5. **Thermal Shock (Ice Bath):** Immediately transfer the eggs using tongs to a bowl of cold tap water. This drastic temperature reduction halts the internal cooking process immediately and causes the albumin to contract slightly from the shell membrane, facilitating flawless peeling.

## Protocol 2: Rehydrating and De-bittering Soya Chunks

Soya chunks require rapid rehydration and thermal treatment to improve palatability and deactivate mild trypsin inhibitors naturally present in raw soy.

1. Boil water in the kettle until the auto-shutoff engages.
2. Place 50 grams of dry soya chunks in a deep, heat-resistant bowl.
3. Pour the boiling water directly over the chunks until fully submerged. Cover the bowl with a plate to trap escaping steam.
4. Allow the chunks to steep for 10 to 15 minutes. The TVP will undergo massive volumetric expansion as it absorbs the water.
5. Drain the excess water, and vigorously squeeze the chunks to expel residual liquid. This vital step removes the inherent bitter soy flavor profile. The chunks are now ready to be consumed, optionally mixed directly into the mess dal for flavor.

## Protocol 3: Sattu Suspension

Sattu is highly hydrophilic but tends to agglomerate into dry lumps if improperly mixed.

1. Add 50 grams of sattu to a shaker bottle or a wide glass.
2. Add a minor volume of room-temperature water to create a highly viscous paste, utilizing a spoon to break down all dry pockets through mechanical shear.
3. Slowly dilute the paste with 300ml to 400ml of cold water.
4. If the budget buffer permits, the addition of rock salt, roasted cumin powder, and a few drops of lemon juice is highly recommended. The ascorbic acid (Vitamin C) from the lemon significantly enhances the bioavailability of the non-heme iron naturally present in the roasted gram.

## The Alternative Matrix: Ultra-Cheap Kettle-Friendly Substitutes

Dietary fatigue is the primary vector for protocol failure in long-term recomposition strategies. Should the subject experience palate fatigue with raw peanuts or sattu, the following

ultra-cheap, kettle-compatible alternatives can be rotated into the ₹50 daily budget without compromising the macronutrient architecture.

## 1. Skimmed Milk Powder (SMP) – The Premium Protein Alternative

For students unable to afford standard whey protein isolate, Skimmed Milk Powder (SMP) serves as a highly concentrated, non-perishable dairy protein source. In India, dairy cooperatives like Amul or Sagar produce spray-dried SMP that is guaranteed to contain a minimum of 35% protein with less than 1.5% fat.

| SMP Alternative Data          | Details                                                                                                                                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cost &amp; Procurement</b> | A 500-gram pouch of Sagar SMP retails for roughly ₹220, or ₹440 per kilogram. Wholesale 1-kilogram pouches can be sourced for ₹290 to ₹340 in local markets.                                                             |
| <b>Nutritional Yield</b>      | Consuming 30 grams of SMP provides roughly 10.5 grams of high-quality casein and whey protein, with virtually zero fat and 15 grams of lactose (zero added sucrose).                                                     |
| <b>Logistics</b>              | SMP dissolves instantly in warm water from the kettle, creating a high-protein milk substitute. It is optimally consumed pre-bedtime, as the casein fraction facilitates a slow, sustained overnight amino acid release. |

## 2. Tofu (Soya Paneer) – The High-Yield Vegetarian Protein

Traditional dairy paneer (averaging ₹250 to ₹400 per kilogram) is a luxury item incompatible with a ₹50/day budget. However, Tofu, commercially marketed in India as "Soya Paneer," is a highly economical, high-protein alternative.

| Tofu Alternative Data         | Details                                                                                                                                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cost &amp; Procurement</b> | In local wholesale markets across Kolkata, fresh soya paneer blocks range from ₹100 to ₹160 per kilogram. Even mid-range retail tofu is priced affordably between ₹180 and ₹220 per kilogram.            |
| <b>Nutritional Yield</b>      | 100 grams of firm tofu provides approximately 14 to 16 grams of protein, roughly 7 to 9 grams of fat, and low carbohydrates, alongside high calcium content derived from the coagulation process.        |
| <b>Logistics</b>              | Tofu is fully cooked during the manufacturing process. It can be safely consumed raw, or cubed and submerged in hot kettle water for 3 minutes to soften and warm before being mixed into the mess rice. |

## 3. Poha (Flattened Rice) – The Rapid Carbohydrate Alternative

While rolled oats are heavily marketed in the fitness industry, they carry an unnecessary

premium. Poha (flattened or beaten rice) is an indigenous, ultra-cheap source of complex carbohydrates that is significantly easier to prepare than oats in a hostel setting.

| Poha Alternative Data         | Details                                                                                                                                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cost &amp; Procurement</b> | Loose white or red poha in Kolkata markets is aggressively priced between ₹40 and ₹65 per kilogram.                                                                                                                             |
| <b>Nutritional Yield</b>      | A 50-gram serving provides robust, easily digestible carbohydrates for pre-workout energy. Red poha is superior, as it retains the bran layer, offering enhanced dietary fiber and a lower glycemic index.                      |
| <b>Logistics</b>              | Unlike raw rice, poha is parboiled and flattened before packaging. It does not require boiling. Placing it in a bowl and covering it with hot water from the kettle for 2 to 3 minutes yields a soft, fluffy carbohydrate base. |

#### 4. Broken Wheat (Dalia) – The High-Fiber Complex Carbohydrate

Dalia is coarsely milled whole raw wheat, exceptionally rich in dietary fiber, B-vitamins, and trace minerals. It is highly satiating and digests at a much slower rate than white rice, providing sustained energy.

| Dalia Alternative Data        | Details                                                                                                                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cost &amp; Procurement</b> | Wheat dalia (from brands like Ganesh or Ankit) retails for a highly economical ₹35 to ₹50 per kilogram.                                                                                                                                              |
| <b>Nutritional Yield</b>      | Dalia offers a highly complex carbohydrate structure. It prevents rapid insulin spikes and supports gut health due to its high insoluble fiber content.                                                                                              |
| <b>Logistics</b>              | While traditional coarse dalia is pressure-cooked, finer grades of dalia can be steeped in boiling water from a kettle. By placing it in a covered thermos or insulated flask with boiling water for 20 minutes, it becomes tender and fully edible. |

## Strategic Chrononutrition and Daily Implementation Schedule

To maximize muscle protein synthesis, total daily protein intake must be evenly distributed across episodic intervals. The human body does not store amino acids in the same manner it stores glycogen or adipose tissue; therefore, frequent spikes of blood amino acid levels are required to maintain an anabolic environment.

### 08:00 AM - Breakfast (Pre-Workout Protocol)

- **Preparation:** Kettle-boil 4 eggs using the passive thermal method. Mix 50 grams of Sattu with cold water.
- **Consumption:** 4 whole eggs + Sattu suspension.

- **Rationale:** The rapid amino acid infusion from the highly bioavailable eggs, combined with the slow-release complex carbohydrates and prebiotic fiber from the sattu, provides sustained ATP generation for the morning training session without causing a hypoglycemic crash mid-workout.

#### 11:00 AM - Lunch (Post-Workout / Mess Meal 1)

- **Procurement:** Mess dining hall.
- **Consumption:** 2.5 standard bowls (katoris) of cooked white rice + 62.5 grams Mess Chicken + 1 bowl watery dal.
- **Rationale:** The high-glycemic white rice rapidly replenishes depleted muscle glycogen stores and spikes insulin. In the post-workout window, insulin serves as a highly anabolic shuttle, driving the amino acids derived from the chicken directly into the damaged muscle tissue to initiate repair.

#### 04:00 PM - Afternoon Protein Synthesis Trigger

- **Preparation:** Rehydrate 50 grams of soya chunks with boiling kettle water.
- **Consumption:** 50 grams Soya chunks + 30 grams Raw Peanuts.
- **Rationale:** This meal delivers a massive 26-gram protein bolus to trigger a secondary MPS spike. The dense, healthy fats from the raw peanuts intentionally slow down the gastric emptying and digestion of the soy protein, ensuring a steady, prolonged trickle of amino acids into the bloodstream during the long afternoon fasting window.

#### 08:00 PM - Dinner (Mess Meal 2)

- **Procurement:** Mess dining hall.
- **Consumption:** 2.5 standard bowls (katoris) of cooked white rice + 62.5 grams Mess Fish (Rohu) + 1 bowl watery dal.
- **Rationale:** The omega-3 rich fish protein aids in overnight recovery, reducing systemic inflammation, while the final carbohydrate load ensures liver glycogen is topped up for the following day's demands.

## Conclusion

Body recomposition within a high-constraint institutional environment is a complex logistical puzzle that demands strict adherence to metabolic fundamentals and retail mathematics. By utilizing local wholesale pricing to construct the ₹43.80 Kettle Stack (Eggs, Soya, Sattu, Peanuts), the subject successfully bridges the severe protein deficiency inherent in standard hostel mess food. Furthermore, by strictly regulating the intake of unlimited mess rice to precisely five standard 150-gram bowls, the subject can lock in a 2,050 kcal deficit without requiring the use of a digital food scale. This protocol demonstrates that advanced sports nutrition and body recomposition are entirely achievable without kitchens, expensive supplements, or significant financial capital, provided the execution remains clinically precise. *This is for informational purposes only. For medical advice or diagnosis, consult a professional.*

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